Speech variation under information-theoretic constraints

- Different information-theoretic (Shannon 1948) correlates are shown to predict different types of variation:
- More informative/unpredictable items maintain higher distinctiveness/are more resilient to variation.
- For example:
  - Frequencies correlate w/ word durations (Baker & Bradlow 2009) & vowel merger rates (Kang, Yoon & Han 2015).
  - Functional load (# of minimal pairs) correlates w/ merger rates & vowel lengths (Wedel et al. 2013).

Shannon, C. E. (1948). A Mathematical Theory of Communication. *Bell System Technical Journal, 27*, 379–423. Baker, R. E., & Bradlow, A. R. (2009). Variability in word duration as a function of probability, speech style, and prosody. *Language and Speech, 52*(4), 391–413. Kang, Y ., Y oon, T. J., & Han, S. (2015). Frequency effects on the vowel length contrast merger in Seoul Korean. *Laboratory Phonology, 6*(3-4), 469–503. Wedel, A., Jackson, S., & Kaplan, A. (2013). Functional load and the lexicon: Evidence that syntactic category and frequency relationships in minimal lemma pairs predict the loss of phoneme contrasts in language change. *Language and Speech, 56*(3), 395–417.

Speech variation under information-theoretic constraints

- How **generalizable** are these findings?
- Most previous studies:

1. Focused on specific types of variation (duration, merger, lenition, etc.): Are more generally applicable (not restricted to specific variation types)?
2. Used acoustic data as an estimate of speech production behavior: Would the effects behave differently w/ regard to acoustic & articulatory dimensions of variation?

Cohen Priva, U. (2017). Informativity and the actuation of lenition. *Language, 93*(3), 569–597.